Coursework Description Sheet

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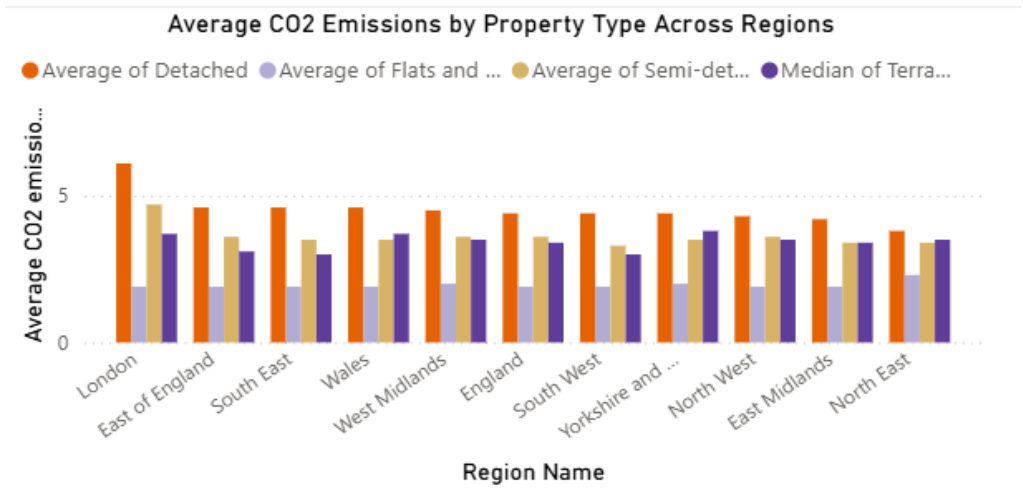
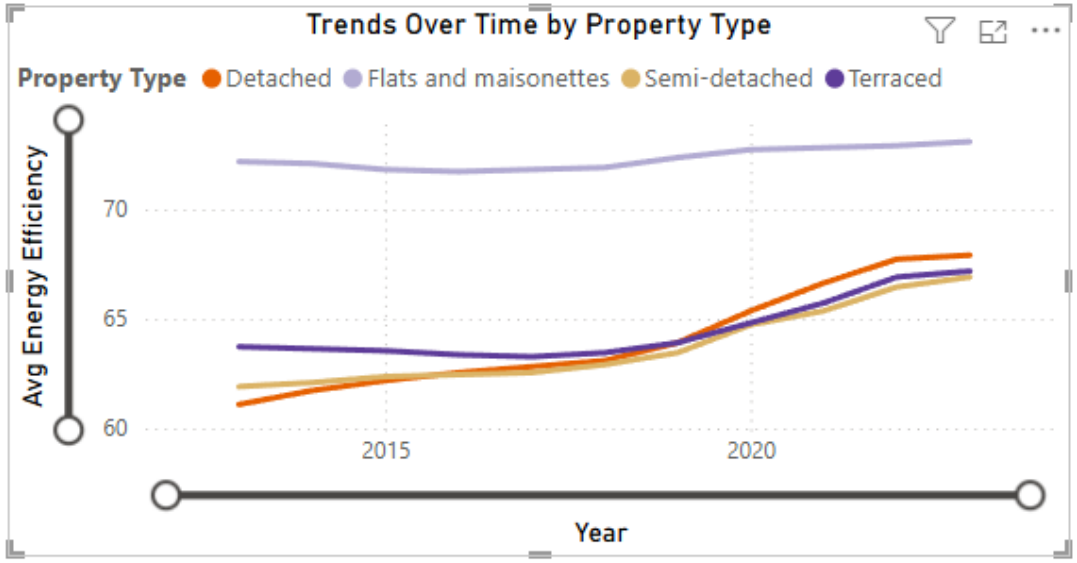
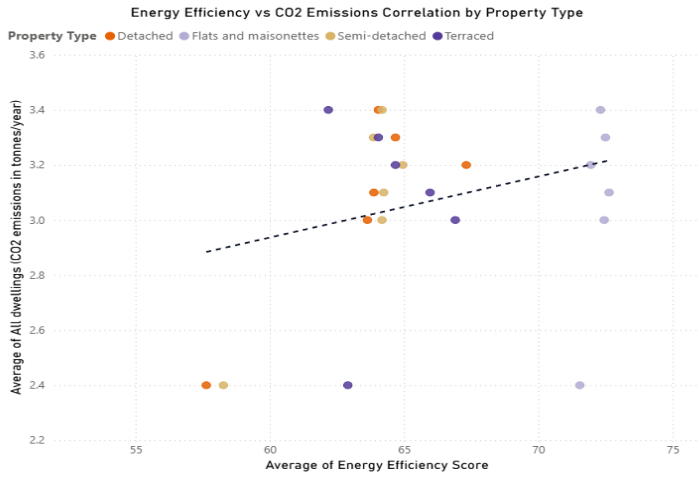
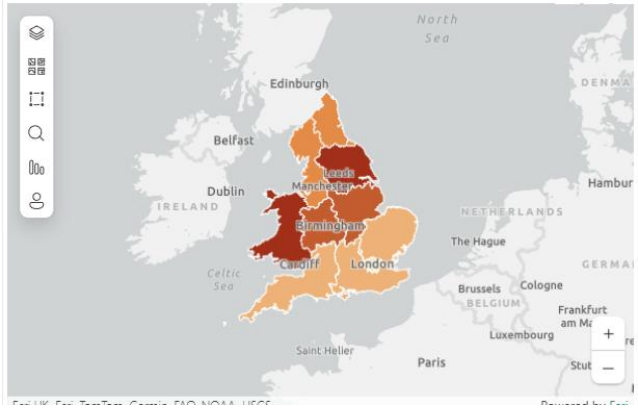
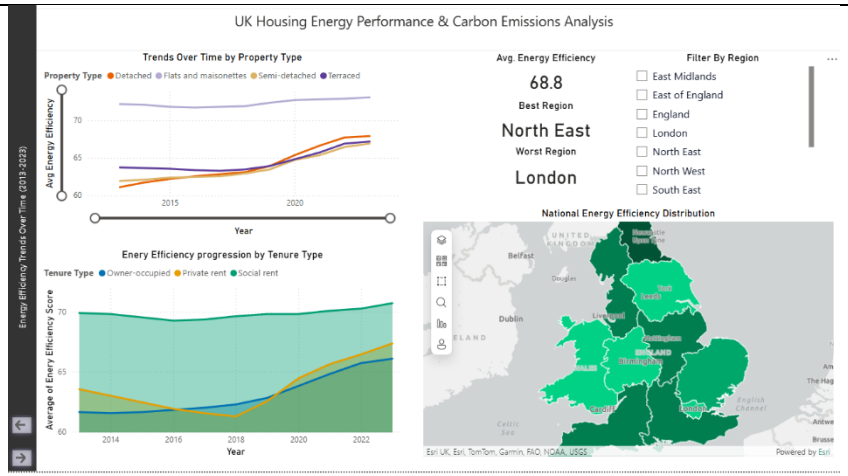
Question	Description	Figure
Fit to Task/User needs		
Location task - How does the visualisation allow users to access the spread of carbon dioxide emission across the UK based on the property type?	To address the location, I attempted to present the data using both bar charts and a UK choropleth map to make the regional CO2 emission patterns intuitive and visually appealing. The dashboard compares emission levels of the different property types, including detached, semi-detached, terraced, and flats/maisonettes. Consider an example of the displayed data, detached properties in London report the highest average emission of around 6.1 tonnes of CO2 per year. In comparison, flats show much lower values (roughly 1.9 tonnes of CO2/year). To help users understand how emissions vary across regions, the map represents each UK region through colour intensity—from lighter shades for low emissions to darker tones where emissions are comparatively high. This design quickly communicates where the environmental "hotspots" lie. The dashboard also features filters by tenure type, which allows users to observe how ownership patterns influence emission levels in different parts of the country. By combining numerical charts with regional mapping, users can analyse both detailed comparisons (like region-to-region differences) and overall national trends. This layered method enhances understanding, and with filters built in, the viewer	

	can easily explore how emissions differ depending on region or tenure type.	
Time task - How does the visualization allow user to understand the evolution of energy efficiency based on the property type, and location?	For the time based part, I used line charts to display the changes in average energy efficiency scores between 2013 and 2023. These visuals highlight steady improvement in property energy efficiency over the years. Flats and maisonettes lead the chart with average scores above 73, while detached and semi-detached houses show progress from roughly 61–62 in 2013 to about 67–68 by 2023. Terraced houses also follow a similar upward pattern. Another visual, a stacked area chart grouped by tenure type, provides additional insights into how ownership affects efficiency trends. Social rent properties maintain high efficiency values (around 70%), while owner-occupied and privately rented dwellings have caught up steadily over the last few years. The dashboard also features a year-range slider, allowing users to adjust the period and view short-term or long-term trends more clearly. Regional filters make it easy to check local performance. For example, the Northeast region exhibits comparatively better energy efficiency, while London remains slightly lower, possibly due to its older housing stock and differing maintenance standards. Overall, these time-series visuals enable users to explore changes and progress in energy performance across both property categories and regions.	

Multi-dimentional data task - How does the visualization allow user to identify correlation amongst at least three of the following parameters: property type, tenure, location, energy efficiency, and carbon dioxide emission?	To represent the multidimensional data task, I have used a scatter plot to reveal relationships between multiple parameters simultaneously—energy efficiency (x-axis), CO2 emissions (y-axis), property type (colour-coded), and location (via interactive filters). This approach enables users to visually discern patterns between the efficiency of a property and the amount of CO2 it produces. From the plot, we can see a clear negative trend: as energy efficiency increases, emissions tend to decrease. The chart includes a trendline summarising that for each 1-point increase in efficiency score CO2 emissions increase slightly. However, these trends differ slightly across regions and property types. Detached houses generally have lower efficiency and higher emissions, while flats cluster in the high-efficiency, low-emission zone. The interactivity is quite essential here. By selecting regions, users can directly see how local conditions or tenure structures influence the relationship. For example, terraced properties in Wales exhibit higher emissions compared to those in London. This makes the dashboard interactive, enabling the target audience to analyse the relationship between CO2 emissions and energy efficiency across various regions.	
Visualisation Principles		
Use of colour - How does the use of colour in this dashboard enhance the readability and effectiveness of the data presentation?	Colours were selected using the ColorBrewer palette to make the visuals clean and accessible even for colourblind viewers. Each property type has its distinct colour scheme—orange for detached, lavender for flats, tan for semi-detached, and navy blue for terraced—which creates consistency and recognisability. The geographic map uses a gradient from light orange to deep brown for emission intensity. This colour strategy makes the dashboard intuitive and visually balanced, aligning with accessibility-friendly design principles by Brewer and Harrower (2003).	

Use of graphic design principles -How does the application of graphic design principles enhance the clarity and effectiveness of the data presentation in this dashboard?	The dashboard layout follows simplicity and visual hierarchy. The main scatter plot, which displays the relationship between energy efficiency and CO2 emissions by property type, is highlighted. Typography, spacing, and the use of white space help minimise distractions. Every element in the dashboard is arranged in a logical reading order, following how users naturally scan pages. Charts related to each concept are placed close together, allowing viewers to connect the ideas easily. These design principles ensure that the end audience understands the story behind the data rather than just getting lost in numbers.	
Use of interaction - How does the use of interactive design elements improve the user's ability to explore and interpret data on this dashboard?	Interactivity is a significant strength of the dashboard. The user can use filters such as Region, Property, and Tenure to explore the dataset in more detail. Instead of static charts, it provides exploratory data analysis—viewers can ask questions like, "Do private rent properties emit more CO2?" or "How is London improving efficiency compared to 2013?". Keeping in all this in mind this dashboard has been designed. The timeline slider adds another interactive dimension for observing trends across years.	
Use of text and legend - How do the use of text and legends contribute to the clarity and user	The text is concise and straightforward, designed to help the target audience interpret visuals effectively without causing any confusion. The font style, size, and colour are kept minimalist. Headings clearly indicate what each chart illustrates, while annotations emphasise key insights. Legends are colour-coded to match property types throughout, ensuring clarity and consistency. The slight correlation statistic note on the scatter plot ("For every 1-	

comprehension of the data presented in this dashboard?	point increase in efficiency score, CO2 emissions increase slightly.”) helps reinforce the data connection quantitatively. Simple, meaningful text supports understanding, which is particularly important for mixed audiences without a technical background.	

References

Add the list of references used in the description in the format of your choice.

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- Office for National Statistics (ONS), updated 18 June 2025, ONS website, methodology, Measuring UK greenhouse gas emissions
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